Comparing Vulnerabilities in Networks: Software Defined Networks in Personal Applications

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1 Introduction

With the ever-changing landscape of the Information Technology (IT) field, it is important to stay up to date on security vulnerabilities and new technologies that are being developed. However, someone who does not work with technology will not be aware of these vulnerabilities or concepts in network security, even though their networks may be the easiest to attack. With the right setup and teaching however, these personal networks can be easily secured. One of the newer technologies in the field that can help towards securing these networks is a Software Defined Network (SDN) controller. These devices allow for a more flexible, efficient, and secure network by separating the control plane which controls the decision making, from the Data Plane which controls the portion of the network that handles the actual data transfer itself. By centralizing the decision making to the SDN controller, it abstracts the underlying hardware and allows for more efficient network management. The goal of this paper is to compare a basic network setup in a personal network to a setup with a SDN controller to show how a person can secure their home network. This comparison is important because it will help to inform people about security vulnerabilities, give them tools to mitigate these vulnerabilities, and create a more secure network. Prior works to this paper, including [1], dive into enterprise applications for this technology, and how it can be used to mitigate large scale attacks. However, this paper will focus on the personal networks that are often overlooked. Towards this goal, VirtualBox will be used to simulate a network and its traffic, while Kali Linux will be used to penetration test the network. The attack that will be performed during the penetration testing is a man in the middle (MITM) attack. The hope for this paper is to alert the average user to security vulnerabilities in their network and give them the tools they need to mitigate them.

2 Background

This section will cover the background information that is necessary to understand the rest of the paper, as well as a home network setup. This includes the different types of networking devices and software tools.

2.1 Networking Devices

- - Router: A device that forwards data between different networks. For example, a home router will be the box from your internet service provider (ISP) that connects your home network to your ISP's network, which will then be forwarded to the internet.
- - Switch: A device that acts like a router, but only forwards data within the same network. Some home setups will have a small switch used to connect multiple devices to Ethernet.
- - SDN Controller: A centralized software entity that manages network switches. It makes routing decisions (control plane) and pushes rules to switches, which only forward packets (data plane). This separates decision-making from packet forwarding.
- - End Device: A device that is connected to the network that can send and/or receive data. This can be a computer, a phone, a printer, television, digital camera, etc.

2.2 Software Tools

These are the software tools that will be used in simulating the network and performing penetration testing.

- - VirtualBox: A software tool developed by Oracle that allows for the creation of virtual machines (VMs). This will be used to create a virtual network and end devices for the simulation. [3]
- - Packet Tracer: A software tool developed by Cisco that allows for the simulation of network traffic and the creation of virtual networks. This will be used to create diagrams of the network topology. [2]

2.3 Attack Type

MITM (Man in the Middle): An attack that involves intercepting and potentially altering communication between two parties. This is the one that will be used in the penetration testing to show the effectiveness of the SDN in mitigating vulnerabilities.

3 Related Works

SDNs are a relatively new technology, but a well-researched one. There are many papers that have been published on the topic of SDNs, and many of them focus on the more intricate details by diving into enterprise applications of the SDN, particularly, on the topic of mitigating large scale attacks. However, prior research has seemingly overlooked home networks, which face different threats and technical constraints. This section reviews relevant literature on SDN security, network vulnerabilities, and simulation tools, and then identifies the specific gap this study addresses.

3.1 SDN for Attack Mitigation

Biswas [1] investigated SDN based mitigation of network attacks like direct denial of service (DDoS), demonstrating how an SDN improved detection and response time of an attack when compared to a traditional network architecture. However, this study exclusively focused on enterprise networks, leaving a gap in understanding of how SDNs can be applied to a home network, where resources are limited and attack vectors differ.

3.2 Traditional Network Vulnerabilities

Rostami et al. [4] examined internet facing router services and found that over 40% of known router internet protocol version 4 (IPv4) addresses had at least one vulnerable service exposed. This failure to follow basic network security practices highlights the vulnerabilities of many networks, particularly home networks, where most users do not have the technical knowledge to secure their networks, or even know that this must be done. This further increases the need for SDNs in home networks, as they can enforce security policies centrally.

In summary, prior research has demonstrated SDN's effectiveness for attack mitigation in enterprise networks [1] and documented widespread vulnerabilities in home routers. However, there seems to be a gap in understanding of how SDNs could affect home network security. This paper aims to fill that gap by simulating those networks before and after the implementation of an SDN and performing penetration testing to show the SDN's effect. The next section will show the methodology for this simulation and testing.

4 Methodology

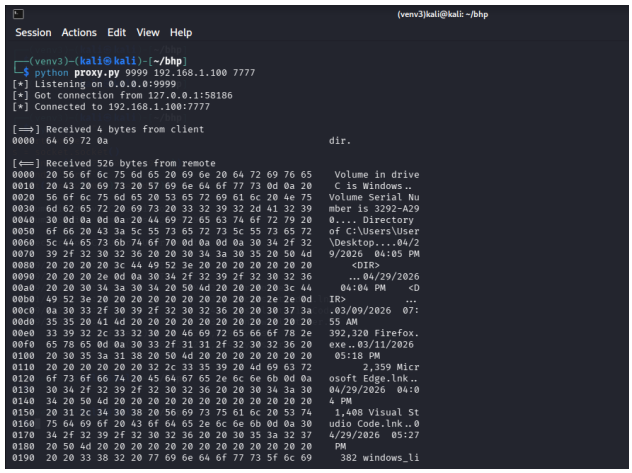
To compare SDNs in enterprise and personal networks, this study uses a simulation-based approach. Through simulating such networks with their traffic, and then performing penetration testing on them, we can see the vulnerabilities of each network, and how the SDN can mitigate those vulnerabilities.

4.1 The Simulation Environment

VirtualBox [3] was used to create virtual machines that represent the different devices on the network. A router was implemented with default settings, in addition to both Windows and Kali Linux machines. The windows machine had a file downloaded to it and executed. This file was a script that performed a MITM attack on the network and was used in tandem with the Kali Linux machine to perform the attack. Through this attack, we can see exactly how vulnerable a local area or personal network can be.

4.2 Home Network with no SDN implemented

The effectiveness of a MITM attack on a home network without an SDN can be shown through the previous diagram of the network topology. Through what would be plugging in one USB drive and running a script on a Linux machine, you have now been compromised, and the attacker can see all of your traffic, and even manipulate it. This was not a difficult



```
(venv3)kali@kali: ~/bhp
Session  Actions  Edit  View  Help

┌──(venv3)─(kali㉿kali)-[~/bhp]
└─$ python proxy.py 9999 192.168.1.100 7777
[*] Listening on 0.0.0.0:9999
[*] Got connection from 127.0.0.1:58186
[*] Connected to 192.168.1.100:7777

[==>] Received 4 bytes from client
0000  64 69 72 0a                                        dir.

[<==] Received 526 bytes from remote
0000  20 56 6f 6c 75 6d 65 20 69 6e 20 64 72 69 76 65    Volume in drive
0010  20 43 20 69 73 20 57 69 6e 64 6f 77 73 0d 0a 20    C is Windows.. 
0020  56 6f 6c 75 6d 65 20 53 65 72 69 61 6c 20 4e 75   Volume Serial Nu
0030  6d 62 65 72 20 69 73 20 33 32 39 32 2d 41 32 39   mber is 3292-A29
0040  30 0d 0a 0d 0a 20 44 69 72 65 63 74 6f 72 79 20   0.... Directory 
0050  6f 66 20 43 3a 5c 55 73 65 72 73 5c 55 73 65 72   of C:\Users\User
0060  5c 44 65 73 6b 74 6f 70 0d 0a 0d 0a 30 34 2f 32   \Desktop....04/2
0070  39 2f 32 30 32 36 20 20 30 34 3a 30 35 20 50 4d   9/2026  04:05 PM
0080  20 20 20 20 3c 44 49 52 3e 20 20 20 20 20 20 20       <DIR>       
0090  20 20 20 2e 0d 0a 30 34 2f 32 39 2f 32 30 32 36      ...04/29/2026
00a0  20 20 30 34 3a 30 34 20 50 4d 20 20 20 20 3c 44    04:04 PM    <D
00b0  49 52 3e 20 20 20 20 20 20 20 20 20 20 20 2e 2e   IR>         ..
00c0  0a 30 33 2f 30 39 2f 32 30 32 36 20 20 30 37 3a   .03/09/2026  07:
00d0  35 35 20 41 4d 20 20 20 20 20 20 20 20 20 20 20   55 AM          
00e0  33 39 32 2c 33 32 30 20 46 69 72 65 66 6f 78 2e   392,320 Firefox.
00f0  65 78 65 0d 0a 30 33 2f 31 31 2f 32 30 32 36 20   exe..03/11/2026 
0100  20 30 35 3a 31 38 20 50 4d 20 20 20 20 20 20 20    05:18 PM       
0110  20 20 20 20 20 20 32 2c 33 35 39 20 4d 69 63 72          2,359 Micr
0120  6f 73 6f 66 74 20 45 64 67 65 2e 6c 6e 6b 0d 0a   osoft Edge.lnk..
0130  30 34 2f 32 39 2f 32 30 32 36 20 20 30 34 3a 30   04/29/2026  04:0
0140  34 20 50 4d 20 20 20 20 20 20 20 20 20 20 20 20   4 PM           
0150  20 31 2c 34 30 38 20 56 69 73 75 61 6c 20 53 74    1,408 Visual St
0160  75 64 69 6f 20 43 6f 64 65 2e 6c 6e 6b 0d 0a 30   udio Code.lnk..0
0170  34 2f 32 39 2f 32 30 32 36 20 20 30 35 3a 32 37   4/29/2026  05:27
0180  20 50 4d 20 20 20 20 20 20 20 20 20 20 20 20 20    PM            
0190  20 20 33 38 32 20 77 69 6e 64 6f 77 73 5f 6c 69      382 windows_li
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Figure 1: This is a screenshot of the data that was captured on the Windows machine during the MITM attack. It was captured after running two commands, one to get the windows machine ready, and the other to pull data onto the Linux Machine. The data that was captured was the output of the directory command, which shows the volume for the drive, the serial number, applications installed, and the raw hexadecimal data.

attack either. There are open resources that allow anyone to learn this type of attack. For example, the resource used for this study was from a small piece of Black Hat Python, [5], a book that is available online for free, and has a chapter dedicated to MITM attacks.

4.3 Mitigation through SDN

Through the use of the SDN controller, we can mitigate the vulnerabilities of the home network. The SDN controller can be used to enforce security policies, such as blocking certain types of traffic, or only allowing certain devices to connect to the network. It also separates the control plane from the data plane, which makes it more difficult for an attacker to gain access to the network, as they would first have to gain access to the far more secure control plane. Below, is the output of the same attack type and code, but with an SDN implemented in the network. As you can see, the attack was not successful, and the attacker was not able to capture any data from the Windows machine. This shows the effectiveness of the SDN in mitigating vulnerabilities in the home network.

5 Conclusion

The insecurities of a home network were brought up throughout this paper. Through the related works and methodology sections this paper described these insecurities, gave resources for taking advantage of them, and showed how an SDN can mitigate a MITM attack. SDN's mitigated these attacks by separating the control and data planes of the network, not allowing certain packets to reach an end user from another device. This is why anyone with a network of their own should take time to look into risks and mitigation techniques, and think about implementing an SDN.

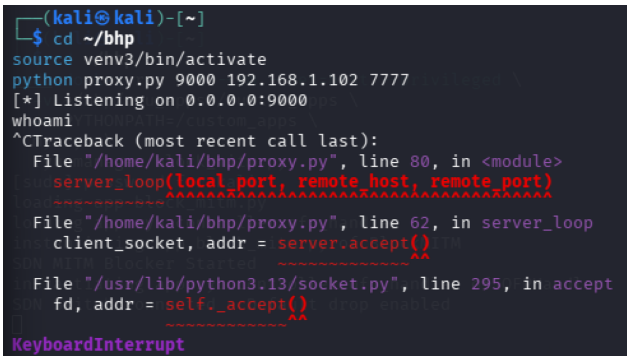

Figure 2: This figure shows the output of the same attack type, same code, same virtual machines, but with the implementation of an SDN in the network. As can be seen, the attack was not successful, and the attacker was not able to capture any data from the Windows machine. This further shows the effectiveness of the SDN in mitigating vulnerabilities in the home network.

6 Addendum: Use of Artificial Intelligence

Generative AI tools were used in a limited and supporting way throughout the writing of this paper. Their use was restricted to brainstorming, clarification of technical concepts, to find sources, and formatting of the Methodology section. At no point was AI used in any way that would be against the Wooster Ethic. When looking at difficult code examples, ChatGPT was occasionally used to describe what the code was doing in the given context, closely relating to how one would use the website Stack Overflow. Most prompts were used to ask technical questions relating to error messages from a Windows terminal.

References

[1] Biswas, R. *Mitigation of Different Network Attacks and Optimization in Software Defined Network*. Ph.d. dissertation, ProQuest Dissertations Publishing, 2021. ProQuest Dissertations & Theses A&I.

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[4] Rostami, S., Heinrich, T., and Albakour, T. Poster: An investigation into internet-facing router services. In *Proceedings of the 2024 ACM on Internet Measurement Conference* (New York, NY, USA, 2024), IMC '24, Association for Computing Machinery, p. 775–776.

[5] Seitz, J., and Arnold, T. *Black Hat Python: Python Programming for Hackers and Pentesters*, 2 ed. No Starch Press, 2021.